NYC Taxi

From ETL to PowerBI

NYC Taxi

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1. Data Overview

The project is based on the official NYC Yellow Taxi Trip Dataset for June 2020 (06/2020). The dataset contains comprehensive record-level information, including precise pickup/dropoff timestamps, trip distances, spatial locations (TLC Taxi Zones), and detailed financial transaction breakdowns. This dataset is utilized to build a full-stack Business Intelligence solution using SSIS for ETL, SSAS Tabular for analytics, and Power BI for interactive reporting.

Data Source

- **Input Format:** CSV file
  Attributes:

- **1. Yellow Taxi Trips Attributes:**
  - **Core & Spatial Features:** `VendorID`, `tpep_pickup_datetime`, `tpep_dropoff_datetime`, `passenger_count`, `trip_distance`, `store_and_fwd_flag`, `PULocationID`, `DOLocationID`.
  - **Rate & Financial Features:** `RatecodeID`, `payment_type`, `fare_amount`, `extra`, `mta_tax`, `tip_amount`, `tolls_amount`, `improvement_surcharge`, `congestion_surcharge`, `airport_fee`, `total_amount`.

- **Taxi Zones Lookup Attributes:**
  - `LocationID`, `Borough`, `Zone`, `service_zone`.

1. Yellow Taxi Trips Attributes (Core & Spatial Fields)

Column Name	Description
VendorID	A code indicating the TPEP provider that provided the record (1 = Creative Mobile Technologies,2 = VeriFone).
tpep_pickup_datetime	The exact date and time when the taximeter was engaged (trip start).

Column Name	Description
tpep_dropoff_datetime	The exact date and time when the taximeter was disengaged (trip end).
passenger_count	The number of passengers in the vehicle, which is a driver-entered value.
trip_distance	The total elapsed trip distance in miles reported by the taximeter.
PULocationID	TLC Taxi Zone ID where the meter was engaged (Pickup Location).
DOLocationID	TLC Taxi Zone ID where the meter was disengaged (Dropoff Location).
store_and_fwd_flag	Indicates whether the trip record was held in vehicle memory before sending to the vendor due to a lack of network connection (Y = store and forward; N = not a store and forward trip)

2. Yellow Taxi Trips Attributes (Rate & Financial Fields)

Column Name	Description
RatecodeID	The final rate code in effect at the end of the trip (1 = Standard rate, 2 = JFK, 3 = Newark, 4 = Nassau or Westchester, 5 = Negotiated fare, 6 = Group ride, 99 = Unknown).

Column Name	Description
payment_type	A numeric code signifying how the passenger paid for the trip (1 = Credit card, 2 = Cash, 3 = No charge, 4 = Dispute, 5 = Unknown, 6 = Voided trip).
fare_amount	The time-and-distance fare calculated by the meter (Base Fare).
extra	Miscellaneous extras and surcharges (such as $0.50 or $1.00 rush hour and overnight charges).
mta_tax	$0.50 MTA tax that is automatically triggered based on the metered rate in use.
tip_amount	The tip amount automatically populated for credit card payments. *(Note: Cash tips are not recorded in this data).*
tolls_amount	Total amount of all highway, bridge, or tunnel tolls paid during the trip.
improvement_surcharge	$0.30 improvement surcharge assessed on trips at the flag drop (introduced in 2015).
congestion_surcharge	Total amount collected for the New York State (NYS) congestion surcharge.
total_amount	The total amount charged to passengers. It includes all fares, taxes, and card tips, but excludes cash tips.

3. Taxi Zones Lookup Attributes

Column Name	Description
LocationID	Unique identifier for each taxi zone, mapping directly to PULocationID and DOLocationID in the trips dataset.
Borough	The name of the New York City borough containing the taxi zone (e.g., Manhattan, Brooklyn, Queens, Bronx, Staten Island).
Zone	The specific neighborhood or area name within the borough.
service_zone	The designated service zone classification assigned by the TLC (e.g., Yellow Zone, Boro Zone, Airports).

Data Quality Handling

- **Negative Values Handled:** Corrected financial anomalies where negative revenue inputs were detected across fare_amount, extra, mta_tax, improvement_surcharge, and total_amount.

- **Zero-Metric Trips Resolved:** Filtered out logical inconsistencies where trip_distance recorded zero miles despite having active trip timestamps

- **Missing Identifiers Fixed:** Resolved NULL values and invalid placeholder codes (like 99) within critical operational and categorical fields including passenger_count, RatecodeID and store_and_fwd_flag using business default fallbacks.

- **Feature Transformation:** Derived a calculated trip duration in minutes (duration_min) by subtracting tpep_pickup_datetime from tpep_dropoff_datetime to add an essential analytical metric to the final dataset.

Analytical Questions

1. What are the peak periods with the highest demand?
2. At what times of day and days of the week is taxi demand at its peak?
3. Is the distribution of demand between Morning and Evening time periods well-balanced?
4. What is the preferred payment method for customers, and does it directly impact revenue and tip volumes?
5. How does weekend demand compare with weekday demand??
6. What is the average tip percentage across different vendors and payment types?
7. Which boroughs hold the longest average trip durations, indicating high transit times?
8. How does each payment method contribute sequentially to our total financial bottom line?

2. Conceptual & Logical & Physical Model

This solution implements a **star schema** design in the Data Warehouse layer for high performance and ease of analysis.

2.1 Conceptual Model

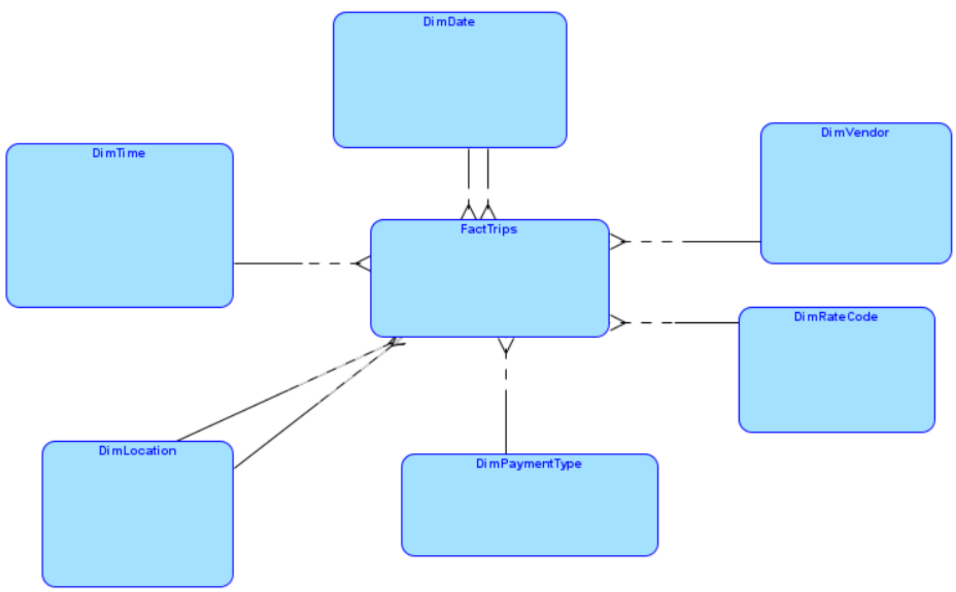
The conceptual model defines the **core entities**, their **roles**, and **relationships** within the business context—abstracted away from technical implementation. It describes **what data exists (Tables)** and **how it's related**, without getting into physical storage or data types.

The conceptual model consists of **one central fact entity** and **six surrounding dimension entities**, forming a **classic star schema** structure.

Entities (Blocks):

- **Fact Trips** *(Fact Table)*
- **Dim Vendor** *(Dimension)*
- **Dim RateCode** *(Dimension)*
- **Dim Payment Type** *(Dimension)*
- **Dim Location** *(Dimension)*
- **Dim Date** *(Dimension)*
- **Dim Time** *(Dimension)*

All dimensions related to fact in 1:M relationship.



2.2 Logical Model

The **logical model** is defined using the following entities:

Dimensions:

• **DimVendor:** Vendor classifications identifying the provider of the trip technology (attributes: VendorKey, VendorID, VendorName).

• **DimRateCode:** Rate code classifications applied to the trips such as standard, JFK, or negotiated fares (attributes: RateCodeKey, RateCodeID, RateCodeName).

• **DimPaymentType:** Payment methods utilized by passengers to settle the trip fare (attributes: PaymentTypeKey, PaymentTypeID, PaymentTypeName).

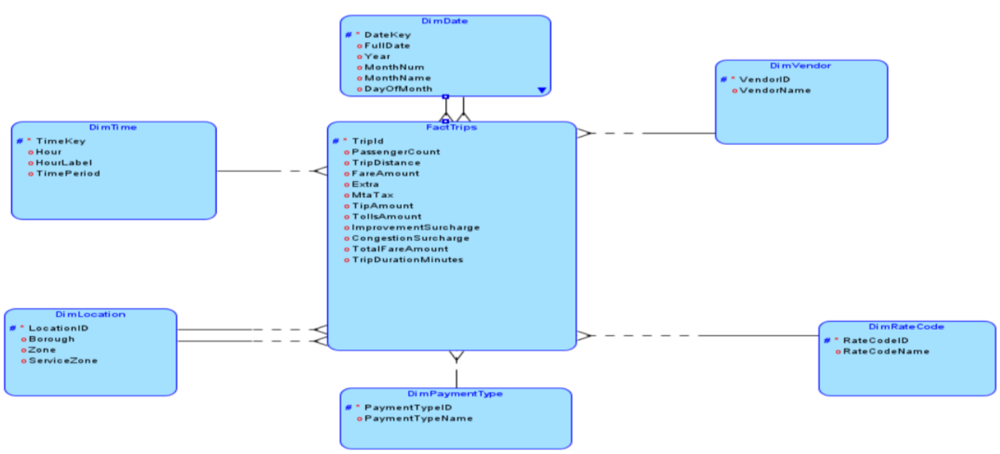
• **DimLocation:** Role-playing dimension representing geographical NYC zones, handling both pickup and dropoff locations (attributes: LocationKey, LocationID, Borough, Zone, ServiceZone).

• **DimDate:** Role-playing dimension handling granular calendar components for trip analysis (attributes: DateKey, FullDate, Year, MonthNum, MonthName, DayOfMonth, DayOfWeek, DayName, Quarter, WeekOfYear, IsWeekend).

• **DimTime:** Role-playing dimension handling specific temporal periods for trip pickup and dropoff times (attributes: TimeKey, Hour, HourLabel, TimePeriod).

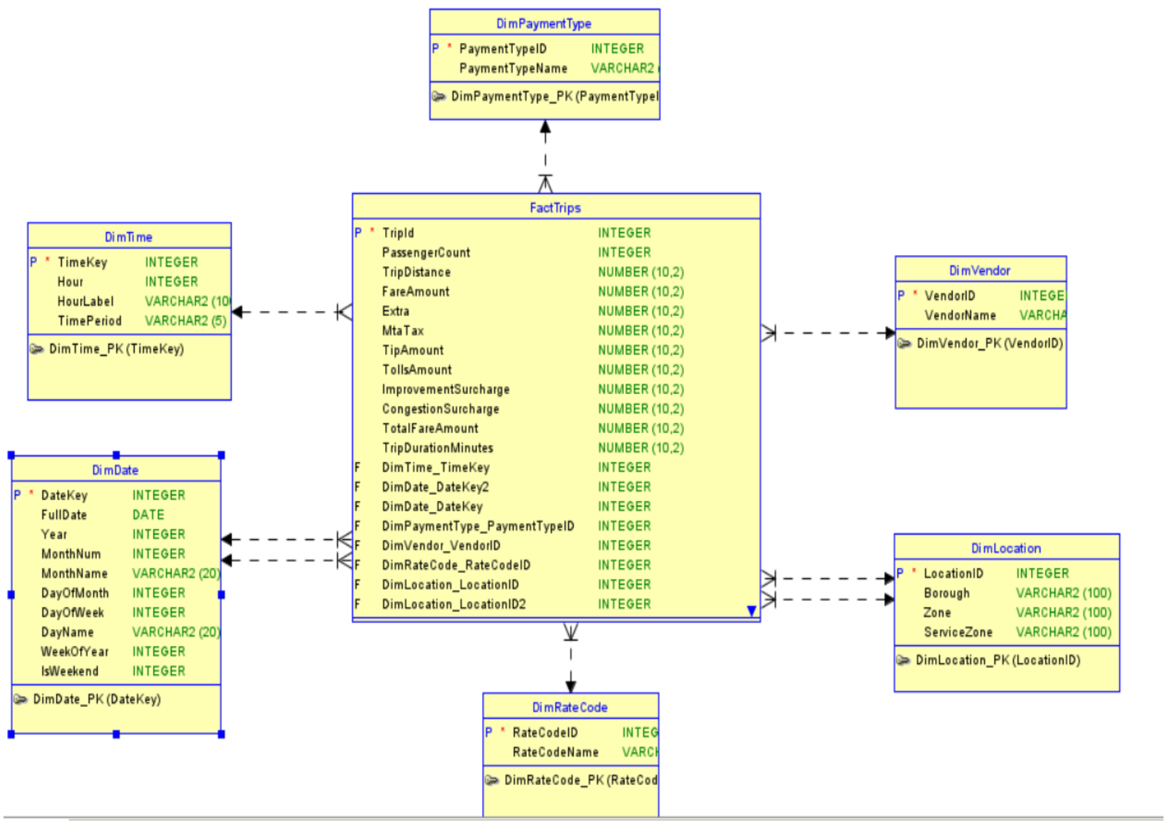
Fact Table:

• **FactTrips:** The central fact table housing measurable operational metrics, financial attributes, calculated values, and trip flags (attributes: PassengerCount, TripDistance, FareAmount, Extra, MtaTax, TipAmount, TollsAmount, ImprovementSurcharge, CongestionSurcharge, TotalFareAmount, TripDurationMinutes, StoreAndFwdFlag).
Each dimension is designed to support one-to-many relationships with the fact table.



2.3 Physical Model

The physical model implements the **logical star schema** within a SQL Server-based Data Warehouse. It focuses on **how data is stored** in the database, **table structures**, and **physical organization**—rather than how the data is loaded**.**



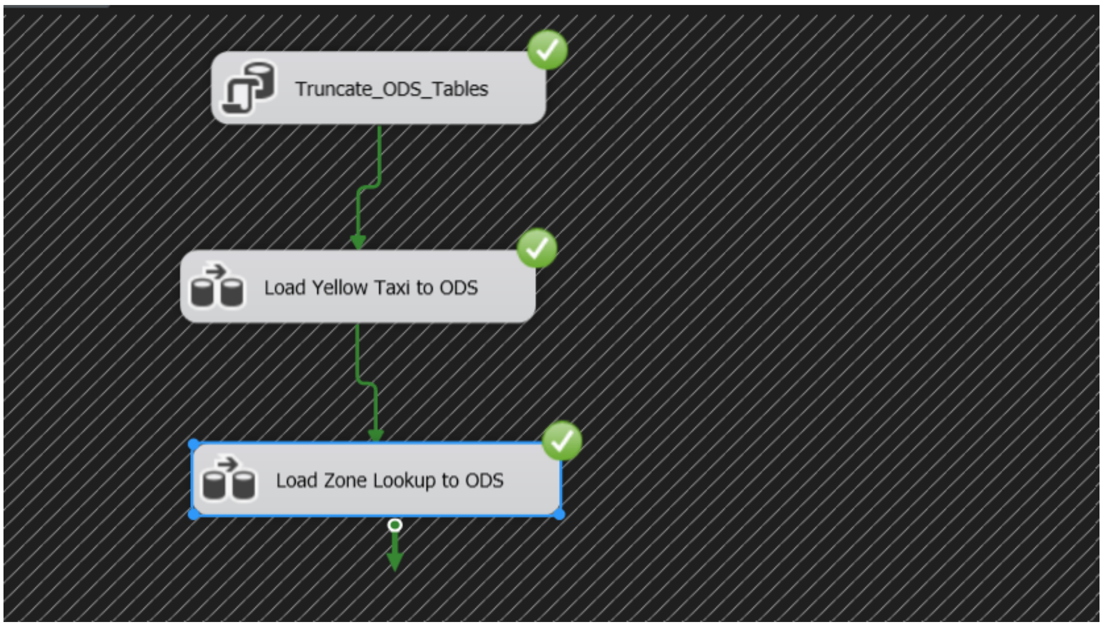
3. SSIS Overview

The SSIS (SQL Server Integration Services) layer orchestrates the **ETL (Extract, Transform, Load)** process in three clearly defined stages, aligned with the architecture's layered design:

ETL Database Layers:

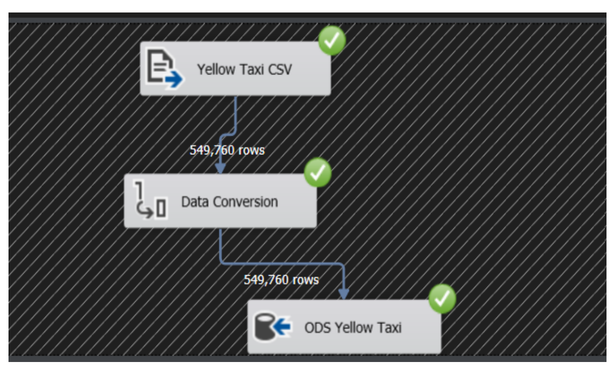
1. **ODS_NYCTaxi**– Raw operational data store
2. **STG_NYCTaxi** – Cleaned and conformed staging layer
3. **DWH_NYCTaxi** – Final dimensional warehouse

A. ODS Package – Operational Data Storage

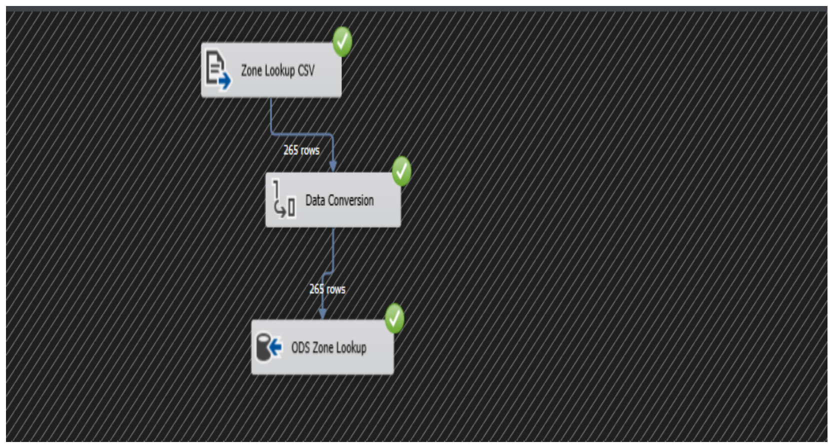


Data Flow Tasks:

1. **DFT – Load Yellow Taxi Data**
    - **Source:** Flat File Source (Yellow Taxi CSV)
    - **Transformation:** Data Conversion
    - **Destination:** OLE DB Destination → ODS_YellowTaxi table
    - **Rows Processed:** 549,760 rows



2. **DFT – Load Zone Lookup Data**
    - **Source:** Flat File Source (Zone Lookup CSV)
    - **Transformation:** Data Conversion
    - **Destination:** OLE DB Destination → ODS_ZoneLookup table
    - **Rows Processed:** 265 rows

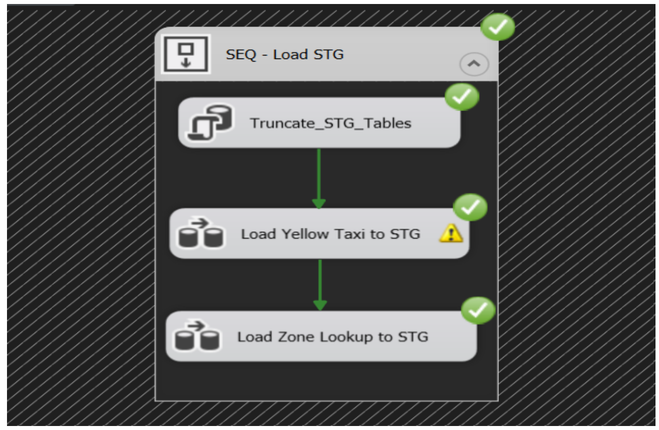


B. STG Package – Staging and Transformation

Objective: Prepare and organize cleaned data into a validated staging layer, applying business rules and data quality checks before loading into the final dimensional warehouse.

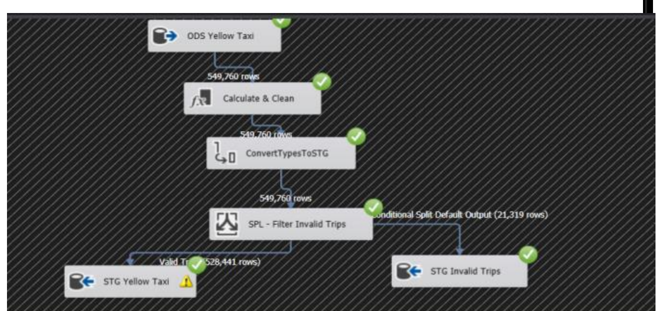
Control Flow:

- Truncate_STG_ Tables
- DFT – Load Yellow Taxi to STG
- DFT – Load Zone Lookup to STG



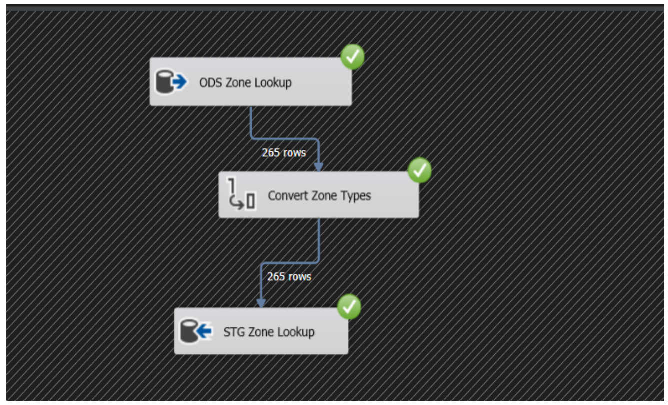
1. DFT – Load Yellow Taxi to STG

- **Source:** ODS Yellow Taxi table (549,760 rows)
- **Transformation 1:** Calculate & Clean — Applied business rules and data cleaning logic, including calculating the trip duration (`duration_min`), handling negative values in financial columns, and resolving invalid placeholder codes (e.g., 99) in fields such as `passenger_count` and `RatecodeID`.
- **Transformation 2:** ConvertTypesToSTG
- **Transformation 3:** SPL – Filter Invalid Trips (Conditional Split)
- **Destination 1:** STG Yellow Taxi table → 528,441 valid rows
- **Destination 2:** STG Invalid Trips table → 21,319 invalid rows



2. DFT – Load Zone Lookup to STG

- **Source:** ODS Zone Lookup table (265 rows)
- **Transformation:**
- **Destination:** STG Zone Lookup table

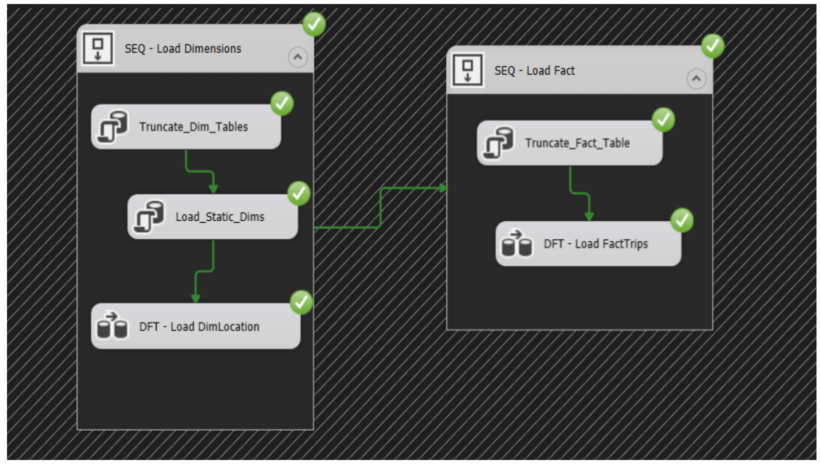


C. DWH Package – Warehouse Load & Lookup

Objective: Populate the dimensional model in DWH_NYCTaxi with clean, fully integrated data from the STG layer, following the star schema design.

Control Flow:

- SEQ – Load Dimensions
- SEQ – Load Fact

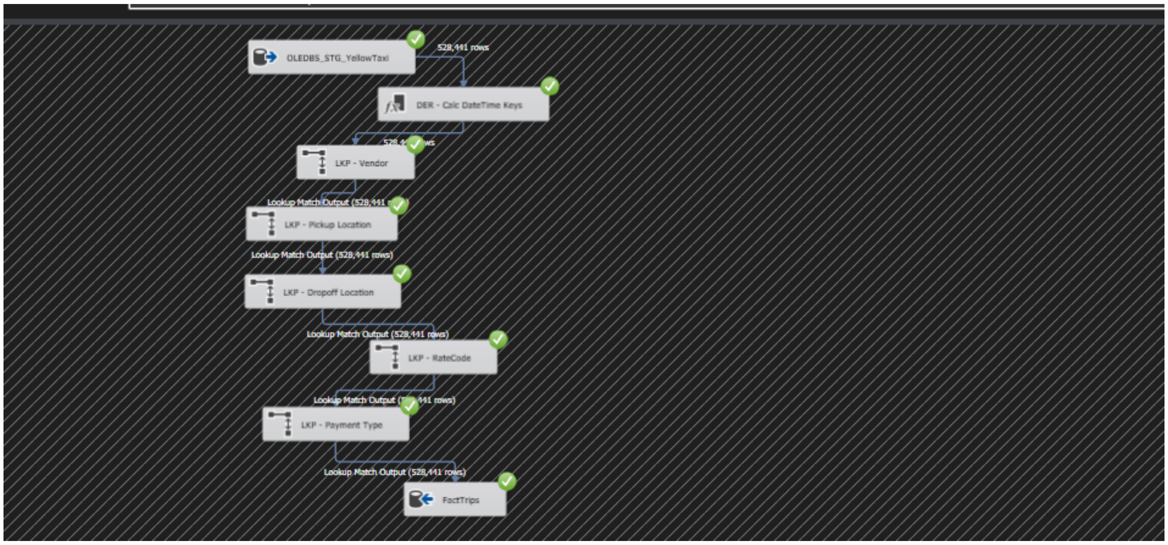


- **1. SEQ – Load Dimensions**

- **Truncate_Dim_Tables**
- **Load_Static_Dims** — Loads the static dimension tables (DimVendor, DimRateCode, DimPaymentType)
- **DFT – Load DimLocation**

2. SEQ – Load Fact

- **Truncate_Fact_Table**
- **DFT – Load FactTrips** — Loads the main fact table, using Lookup Transformations to match Foreign Keys (such as VendorKey, RateCodeKey, PaymentTypeKey, and LocationKey for both Pickup and Dropoff) against the previously loaded dimension tables.


OLEDBS_STG_YellowTaxi
528,441 rows
DER - Calc DateTime Keys
LKP - Vendor
Lookup Match Output (528,441 rows)
LKP - Pickup Location
Lookup Match Output (528,441 rows)
LKP - Dropoff Location
Lookup Match Output (528,441 rows)
LKP - RateCode
LKP - Payment Type
Lookup Match Output (528,441 rows)
FactTrips

4. SSAS Overview

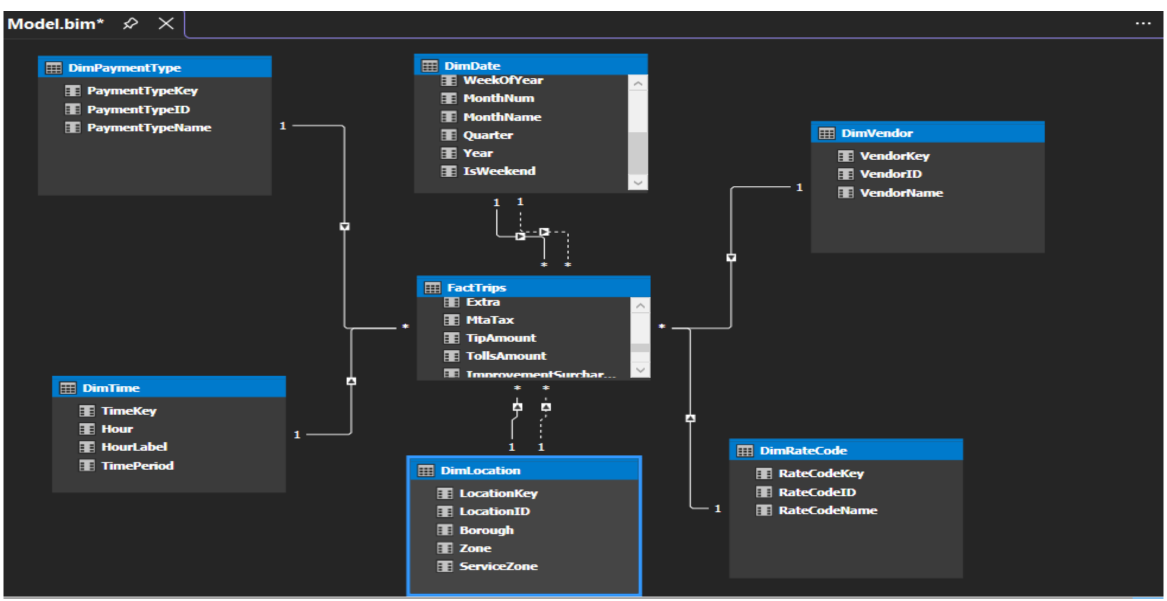
Objective: the NYC Yellow Taxi (June 2020) dataset was modeled using **SQL Server Analysis Services (SSAS) Tabular Model**, enabling fast in-memory analytics through the xVelocity storage engine and DAX for calculations.

After loading the data from the DWH layer, relationships were defined between the **FactTrips** table and its surrounding dimensions (**DimDate, DimTime, DimLocation, DimVendor, DimRateCode, DimPaymentType**) to form a structured star schema semantic model.

Key Modeling Note — Role-Playing Dimensions:

Both **DimLocation** and **DimDate** (along with **DimTime**) are used as **role-playing dimensions** in this model, since each trip has two distinct contexts: **Pickup** and **Dropoff**.

- **DimLocation:** Linked twice to FactTrips (Pickup Location and Dropoff Location). Only one relationship is set as **Active**, while the second is **Inactive**.
- **DimDate:** Linked twice to FactTrips (Pickup Date and Dropoff Date). Similarly, only one relationship is **Active**, with the second defined as **Inactive**.For all TWO dimensions, the inactive relationship is activated at query time using the `USERELATIONSHIP` DAX function inside dropoff-specific measures.
- *"Since the Fact table has multiple relationships with `DimLocation` and `DimDate` (for both Pickup and Dropoff), the Dropoff relationships are kept **inactive** by default. They are dynamically activated at query time using the `USERELATIONSHIP` DAX function inside specific Dropoff measures."*



- Measures written in **DAX**

[DateKey]

	DateKey	FullDate	DayOfWeek	DayName	DayOfMonth	WeekOfYear	MonthNum	MonthName	Quarter	Year	IsWeekend
2	20200101	2020-01-01	4	Wednesday	1	1	1	January	1	2020	FALSE
3	20200102	2020-01-02	5	Thursday	2	1	1	January	1	2020	FALSE
4	20200103	2020-01-03	6	Friday	3	1	1	January	1	2020	FALSE

Total Trips: 528,441	Weekend Trips: 102,881	Total Extra: $554,703.05	Trips With Tip: 320,573	Trips by Zone: 528,441	Avg Trip Duration: 14.4
Total Revenue: $9,920,834.63	Weekday Trips: 425,560	Total MTA Tax: $262,702.85	% Trips With Tip: 60.7%	Revenue by Borough: $9,920,834.63	Avg Speed mph: 17.8
Total Fare Amount: $7,133,559.35	% Weekend Trips: 19.5%	Total Tolls: $192,112.83	Avg Tip When Tipped: $2.97	Avg Trip Duration by Borough: 14.4	Avg Passenger Count: 1.3
Avg Fare per Trip: $13.50	Avg Revenue per Day: 27106.1055464481	Total Improvement Surcharge: $158,492.10	Avg Tip Percentage: 13.4%	Trips by Dropoff Zone: 528,441	
Total Distance: 2,248,199.62		Total Congestion Surcharge: $1,073,645.00		Revenue by Dropoff Borough: $9,920,834.63	
Avg Trip Distance: 4.25		Total Tips: $952,737.60			
		Calculated Total Check: $10,327,952.78			
		Tip Percentage: 13.4%			

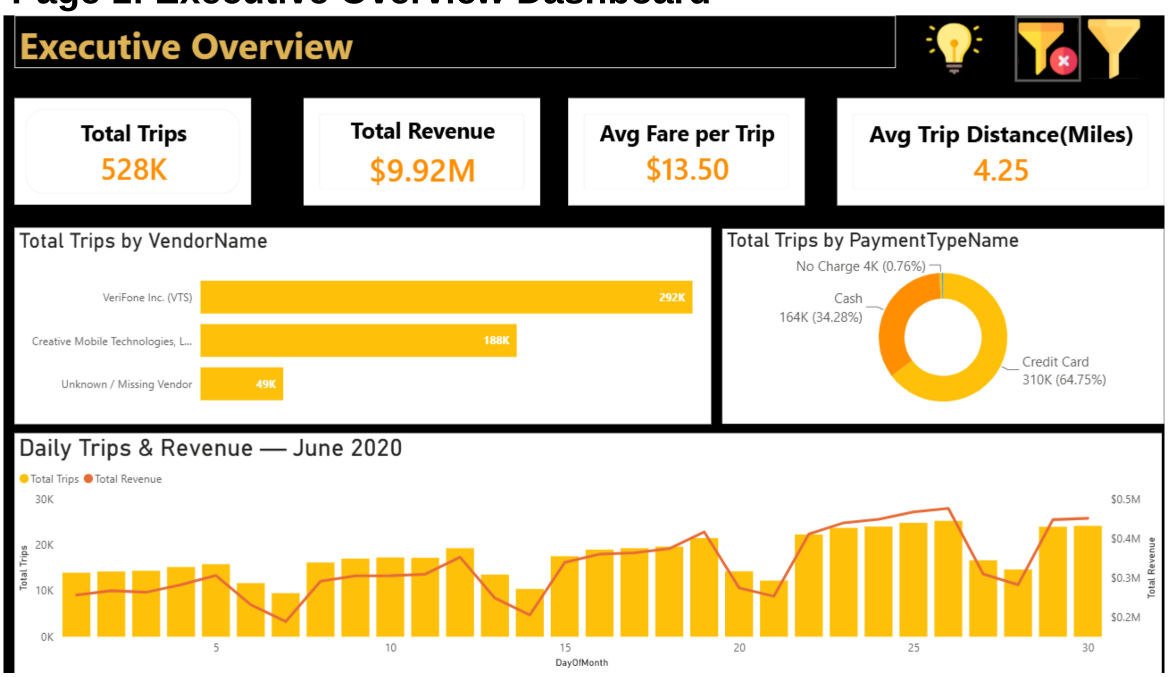
DimDate | DimLocation | DimPaymentType | DimRateCode | DimTime | DimVendor | FactTrips

5. Power BI Overview

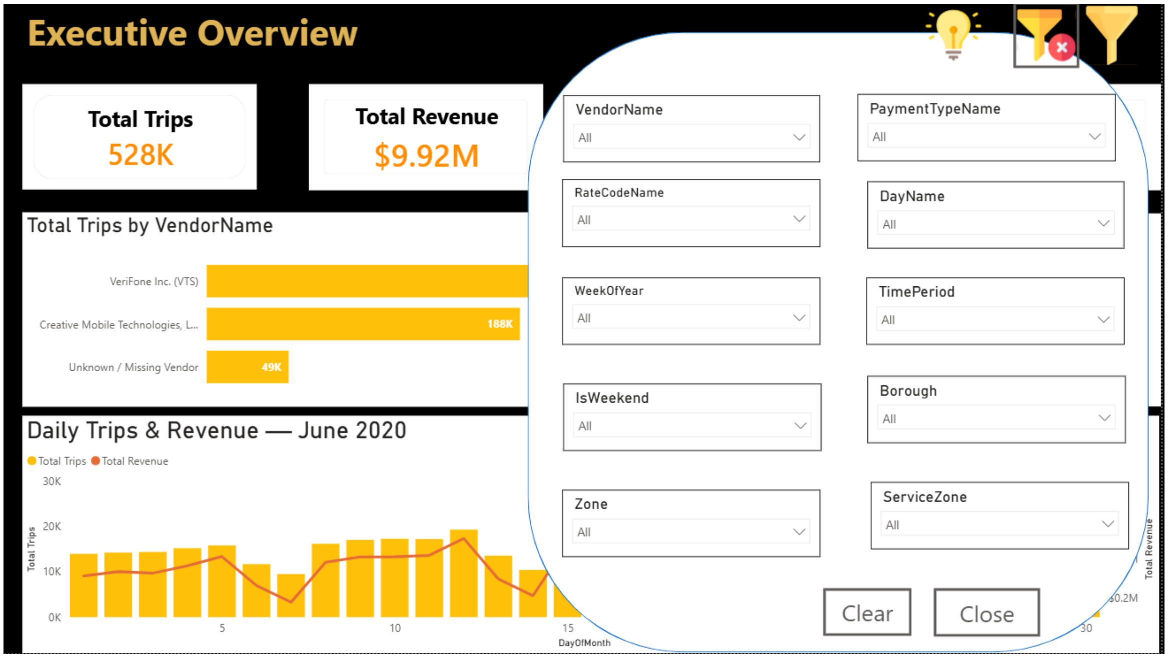
To visualize and analyze the cleaned and structured NYC TAXI data, a dynamic and interactive **Power BI report** was developed. This report connects directly to the **Data Warehouse** loaded via SSIS.

The Power BI report is structured into **four main pages**, each designed to focus on a specific area of analysis:

Page 1: Executive Overview Dashboard

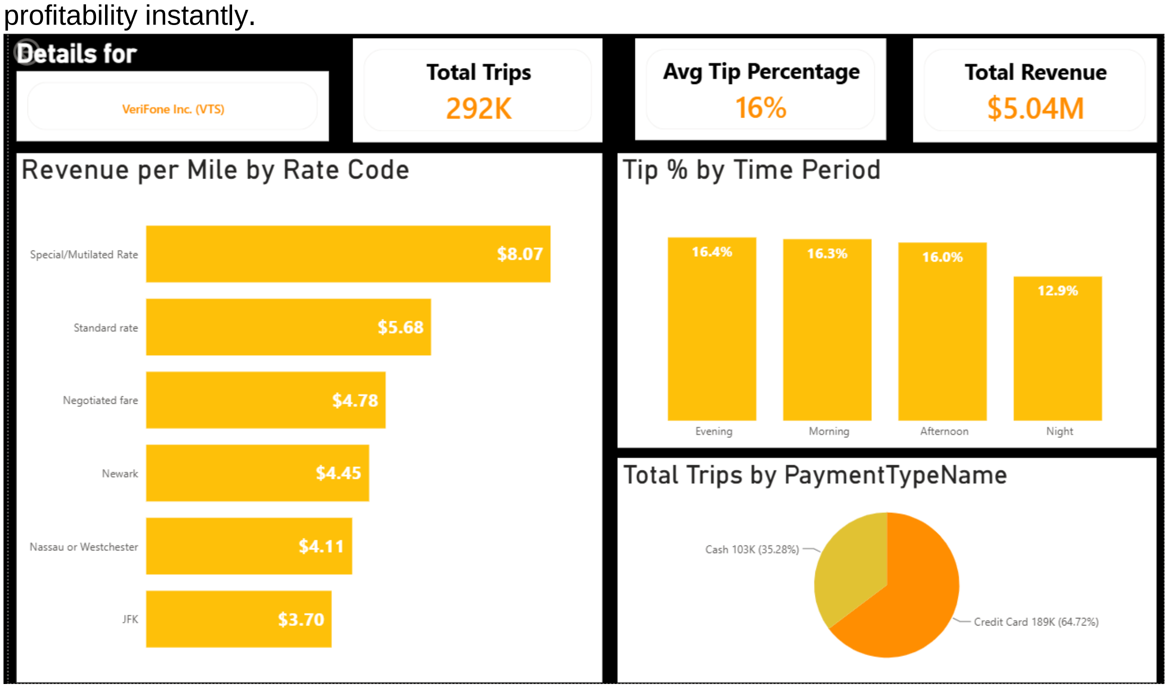


Filter Panel: Users can filter the entire view by specific dates or dimensions to see localized Performance

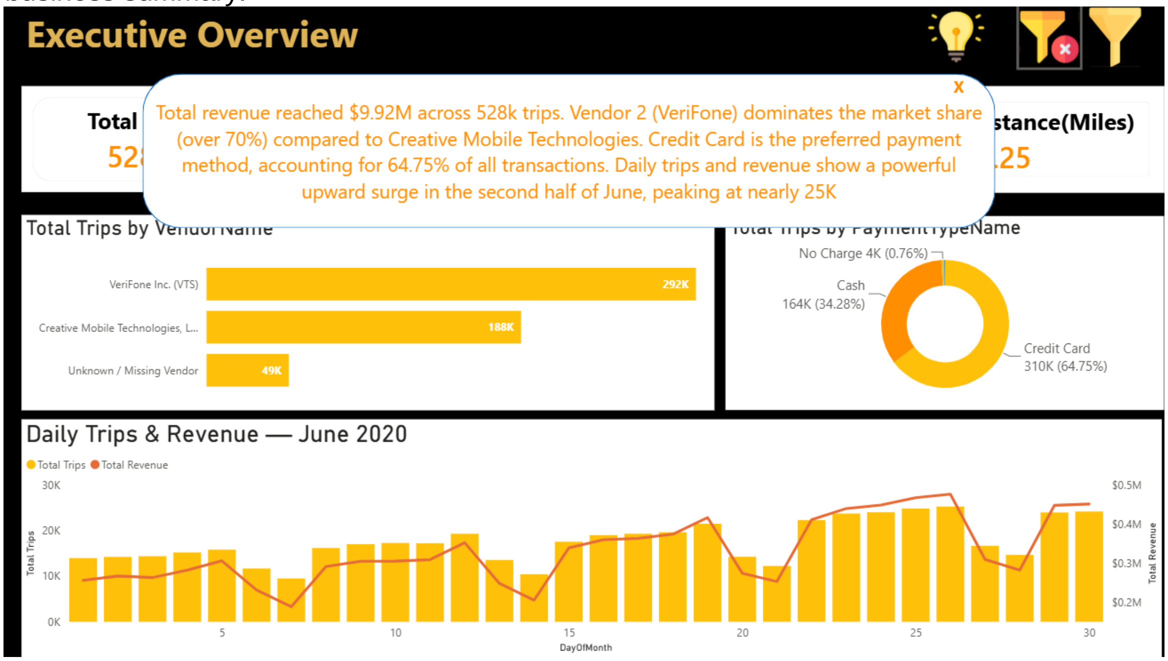


Drill-Through: Vendor Performance Deep-Dive:

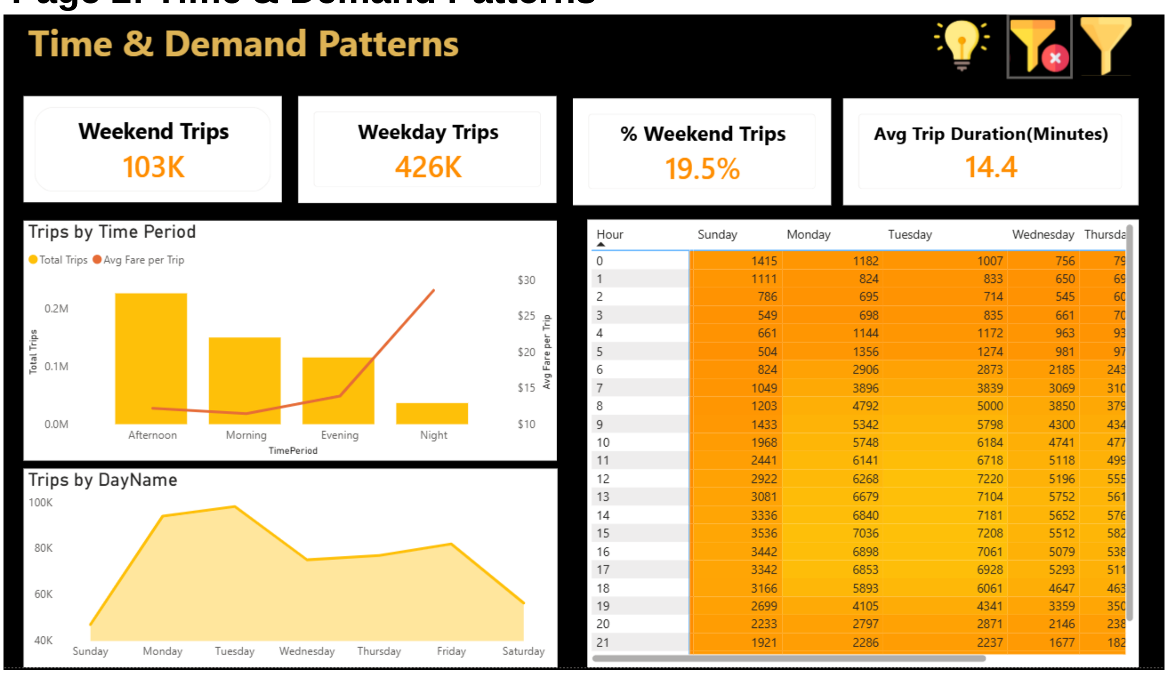
Right-click any vendor on the Executive Overview to unlock an operational efficiency dashboard.
This seamless transition filters the **Revenue per Mile**, tracks **Tip Percentages**, and uncovers **Preferred Payment Methods**—helping managers compare VeriFone and Creative Mobile's profitability instantly.



Insights Toggle : Clicking this button triggers a pop-up window containing the quick business summary.

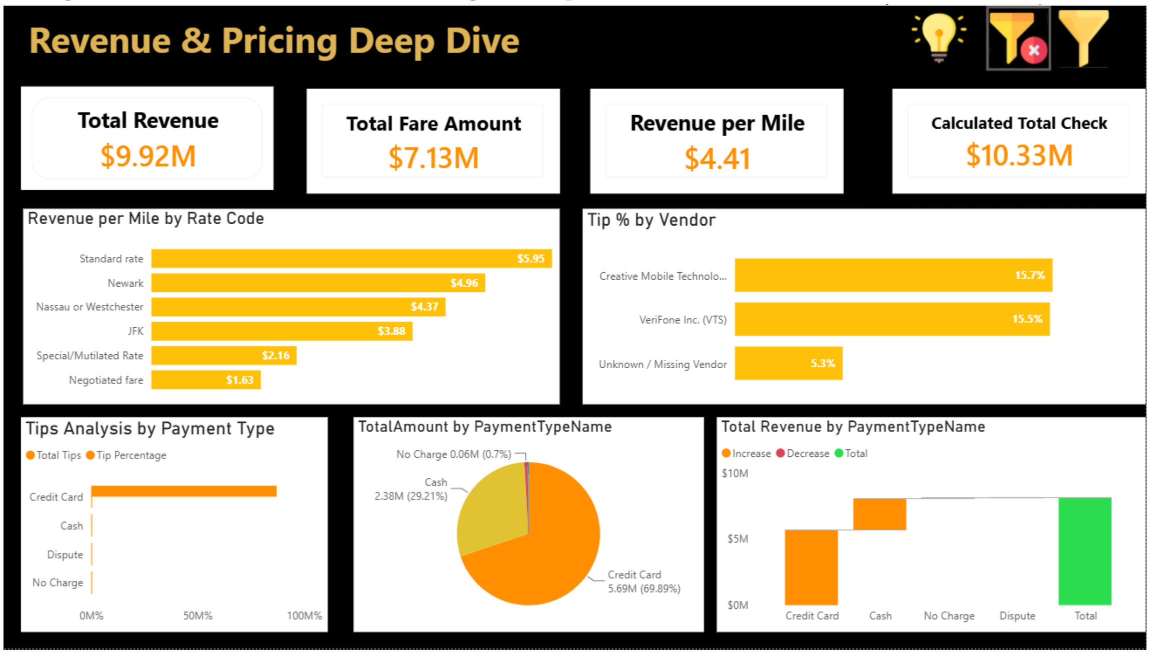


Page 2: Time & Demand Patterns

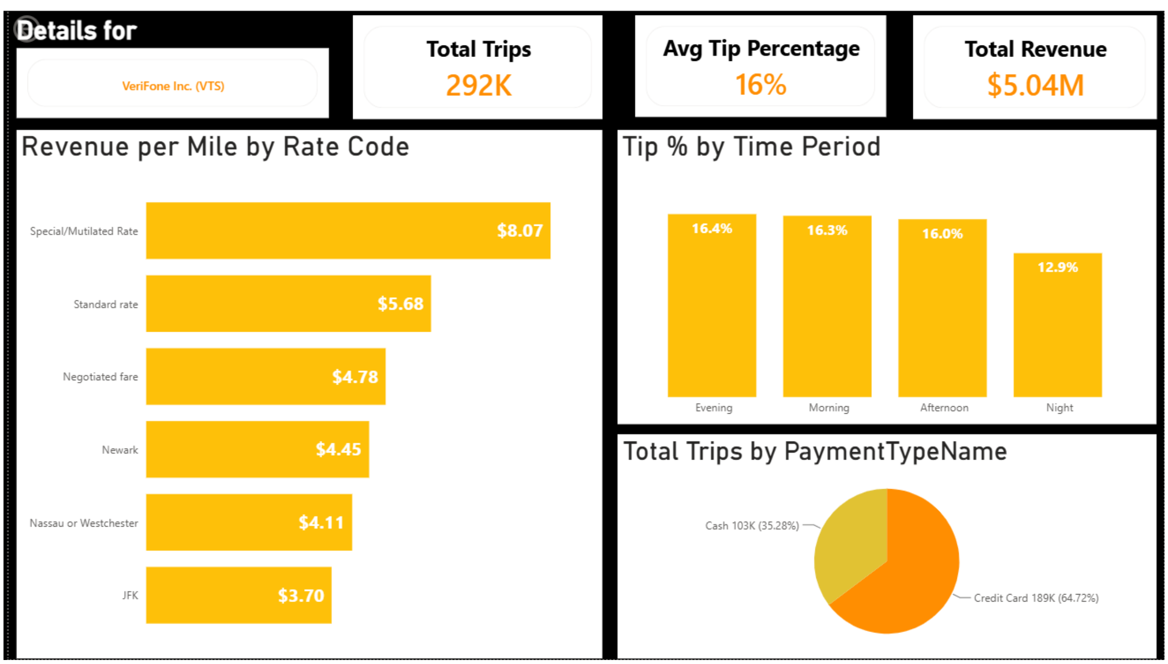


Hour	Sunday	Monday	Tuesday	Wednesday
0	1415	1182	1007	756
1	1111	824	833	650
2	786	695	714	545
3	549	698	835	661
4	661	1144	1172	963
5	504	1356	1274	981
6	824	2906	2873	2185
7	1049	3896	3839	3069
8	1203	4792	5000	3850
9	1433	5342	5798	4300
10	1968	5748	6184	4741
11	2441	6141	6718	5118
12	2922	6268	7220	5196
13	3081	6679	7104	5752
14	3336	6840	7181	5652
15	3536	7036	7208	5512
16	3442	6898	7061	5079
17	3342	6853	6928	5293
18	3166	5893	6061	4647
19	2699	4105	4341	3359
20	2233	2797	2871	2146
21	1921	2286	2237	1677

Page 3: Revenue & Pricing Deep



Drill-Through: Vendor Performance Deep-Dive



Page 4: Geography

Drill-Through: From Borough Overview to Zone

The Geography page displays a dynamic tree-map of trip distribution across NYC boroughs. To prevent information clutter